

Project proposal

Problem statement

For my project I will be implementing conway's 'Game of Life'. Game of Life cellular automaton simulation that takes an initial state and then modifies every cell each tick based on a set of rules and the 8 cells around it.

Language selection

I will implement the Game of Life in the following languages:

1. LUA
2. JULIA
3. Haskell

I choose these three languages because I don't think inform7 is a suitable language for cellular automaton simulation. I will however try to do an implementation with it if I have time because I wanna see if its possible.

Evaluation Criteria

I will evaluate each languages on the following criteria:

1. Memory_usage (objective)
 1. For each language I will track total and average memory usage on the same seed.
2. Frame_duration (objective)
 1. For each language I will track the average time taken to generate each frame using the same seed for each.
3. Developer Experience (subjective)
 1. I will give my thoughts on how it felt to program the Game of Life in each language and why I would or would not recommend the language.